

CASEY HACKETT

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WORK EXPERIENCE

Volos - Systematic Options Trading Tools

June 2024 - Present

- Develop machine learning models to process high-frequency financial data and extract actionable signals.
- Build automated pipelines to clean, normalize, and structure datasets for downstream modeling.
- Collaborate across teams to align business objectives and ensure the delivery of data-driven solutions.
- Implement software automation tools to improve production efficiency and reduce manual intervention.

BTCS – DeFi Arbitrage Research

Jan 2025 – Present

- Design algorithms to detect real-time inefficiencies in cross-DEX markets.
- Create simulation environments to benchmark arbitrage strategies under noisy and high-speed data conditions.
- Conduct machine learning analysis on token price and volume feeds, developing predictive models to identify recurring inefficiencies and inform strategy.

MIT – Quantum Computing Process Optimization

May 2023 – Aug. 2023

- Automated calibration workflows for superconducting qubits using Python and novel optimization routines.
- Streamlined data acquisition pipelines and adaptive control parameters with experimental physicists.
- Contributed to low-level testing of quantum hardware interfaces, including FPGA debugging and signal mapping.

PROJECTS

First Base Defense CV Prototype – Baseball Analytics

2025

- Build a computer vision pipeline (YOLOv8 + DeepSORT) to extract pose and ball-tracking signals from game footage.
- Integrate CV-derived features with play-level defensive metrics (Statcast, RE24) to test predictive value in player evaluation models.
- Outline a roadmap for scaling CV research into broader baseball analytics workflows.
- Experiment with methods to quantify the predictive value of CV-derived features for defensive evaluation models.

Carolina Hurricanes Defense CV Analysis – Hockey Analytics

2023

- Implemented a CV system to quantify defensive positioning and skating explosiveness in transition plays.
- Combined pose and movement mechanics signals with event-level data for performance prediction.
- Explored scalable methods for incorporating CV-derived features into predictive frameworks used by hockey analysts.
- Produced a detailed analysis report demonstrating how CV-derived features could enhance predictive hockey models.

EDUCATION

Lehigh University

Aug 2024 - Present

MS in Financial Engineering

Expected Graduation: May 2026

Relevant Coursework: Statistical Machine Learning, Stochastic Modeling, Data Mining, Optimization

University of Maryland, College Park

Aug 2020 - May 2024

Bachelor of Computer Science: Quantum Information

Physics Minor

TECHNICAL SKILLS

Languages Python, R, Rust, OCaml, C, C++, C#, SQL

Libraries Pandas, NumPy, Scikit-learn, TensorFlow, OpenCV, PyTorch, LightGBM, XGBoost